

Update on 2026 tracking efficiencies (and 2024 systematics)

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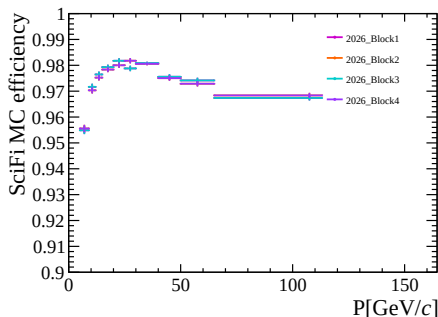
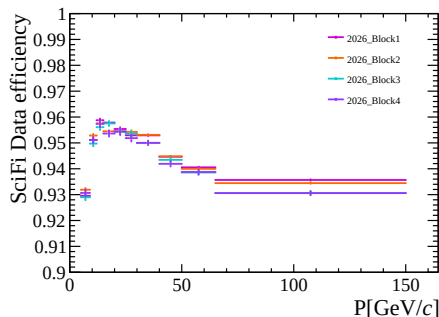
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Progress update:

- ① 2026 efficiencies calculated in 1D using expected MC
 - 2D tables could easily be produced...
 - ... but is this needed?
- ② 2024 analysis:
 - ✓ Data/MC tupling of 24r2 successful
 - ✓ Reweighting working
 - Need to retuple data with CONDITIONAL runs
 - Systematics considered: Multiplicity reweighting, mass shape, Combined-MuonUT discrepancy
 - Separate tables for μ^+ , μ^- , $\mu^\pm$

2026 tracking efficiencies

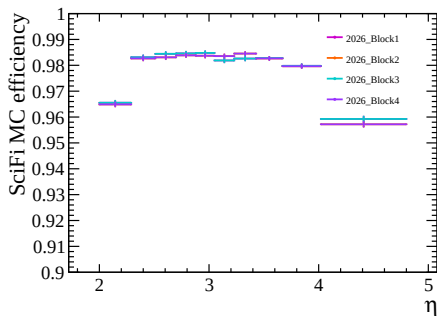
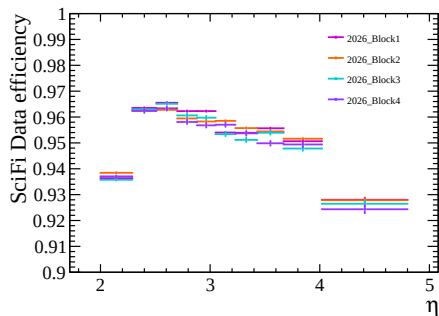
Comparison of tracking efficiencies in 2026 data blocks



SciFi efficiencies looks quite different between data and MC...

2026 tracking efficiencies

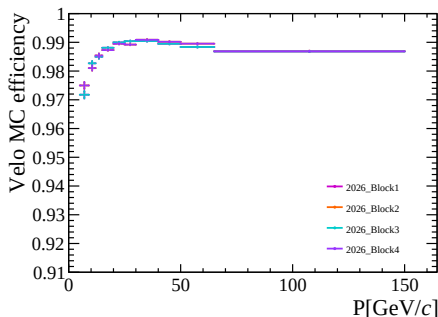
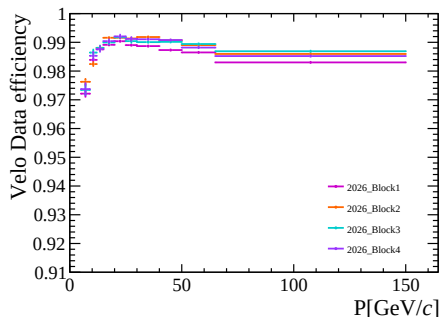
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2026 tracking efficiencies

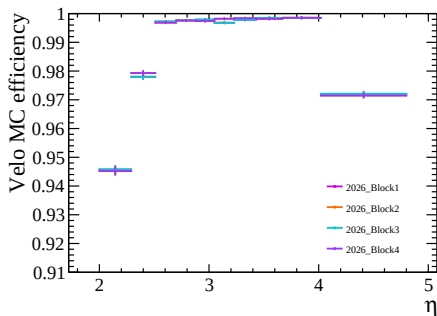
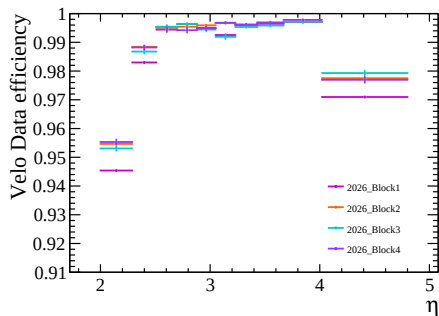
Comparison of tracking efficiencies in 2026 data blocks



Velo efficiencies agree reasonably well between data and MC...

2026 tracking efficiencies

Comparison of tracking efficiencies in 2026 data blocks



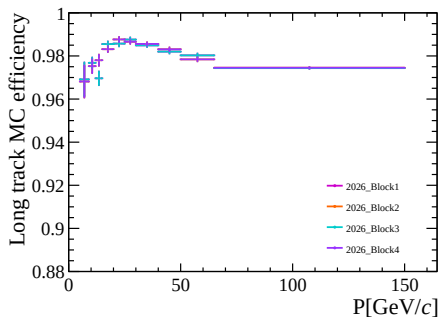
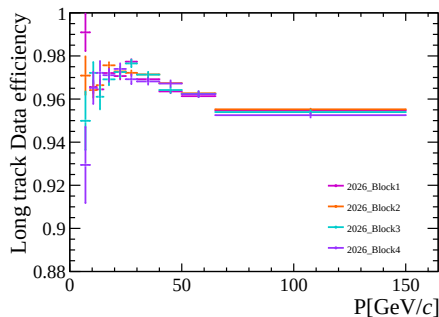
Velo efficiencies agree reasonably well between data and MC...

Velo changes during data taking

- It's unclear why the Velo efficiencies are higher in Blocks 2, 3, 4
- From Run meeting Velo minutes on 20th March 2026:
 - "Tuesday. HV recipes updated. STANDBY2 with 400V for inner sensors and 200V for outer sensors. M30 VP2 with 600V at ready"
- From Run meeting Velo minutes on 24th March 2026:
 - "HV recipes updated with M30/M31 VP2 and VP0 at 600V"
 - "... After some investigation, M30 VP2-0 was confirmed to be damaged. HV recipes rollback to 400V inner sensors and 200V outer sensors."
- Magnet flip happened on 25th March 2026
- Clearly there were changes to HV settings of the Velo during this period, but it's unclear if this can explain the increase in Velo efficiency at high and low η , especially since around this period the first failed Velo modules started appearing

2026 tracking efficiencies

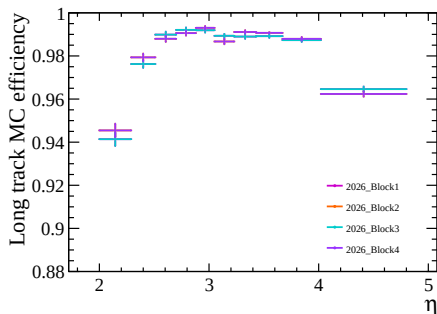
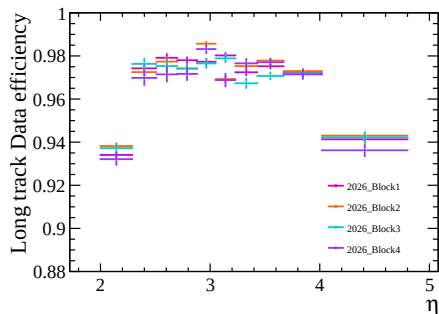
Comparison of tracking efficiencies in 2026 data blocks



MuonUT tracking efficiencies have consistent trend between data and MC

2026 tracking efficiencies

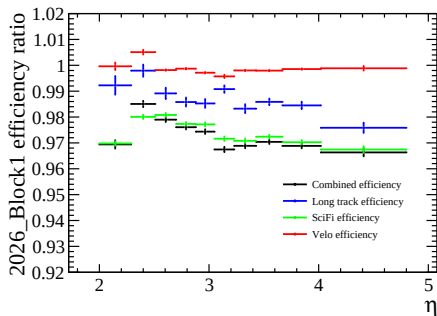
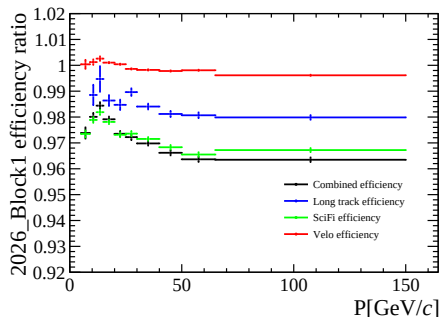
Comparison of tracking efficiencies in 2026 data blocks



MuonUT efficiencies have consistent trend between data and MC

2026 tracking efficiency ratios

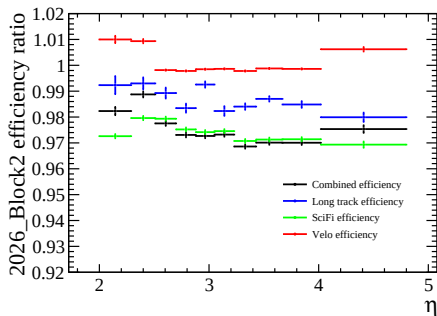
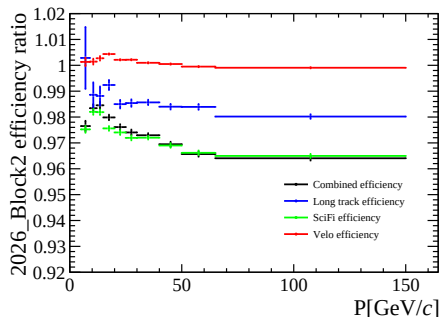
Comparison of tracking efficiency ratios in 2026 Block 1



No surprises in 2026 Block 1

2026 tracking efficiency ratios

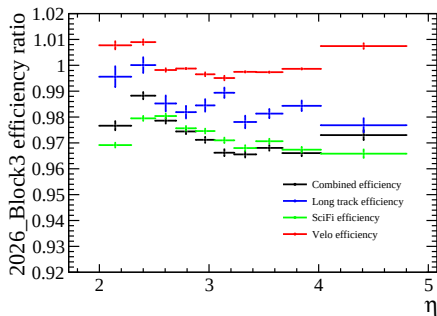
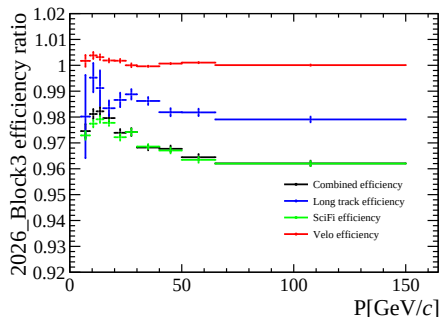
Comparison of tracking efficiency ratios in 2026 Block 2



2026 Block 2 shows some discrepancies in Velo efficiencies

2026 tracking efficiency ratios

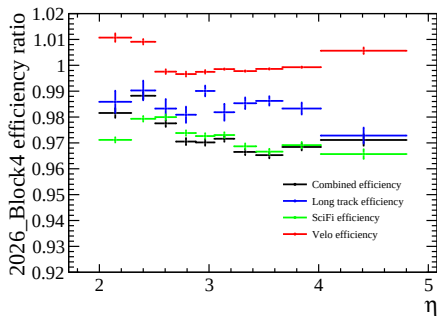
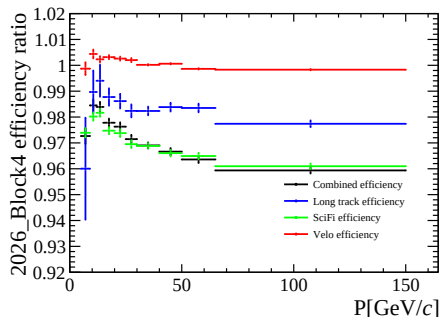
Comparison of tracking efficiency ratios in 2026 Block 3



2026 Block 3 shows some discrepancies in Velo efficiencies

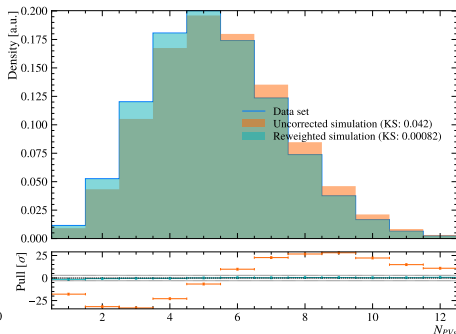
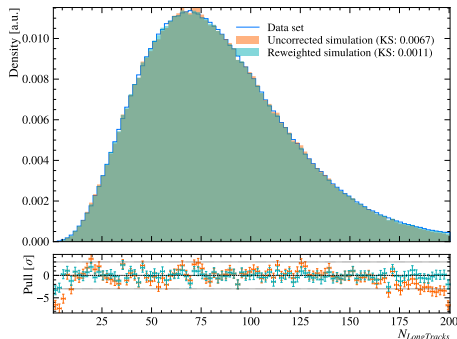
2026 tracking efficiency ratios

Comparison of tracking efficiency ratios in 2026 Block 4



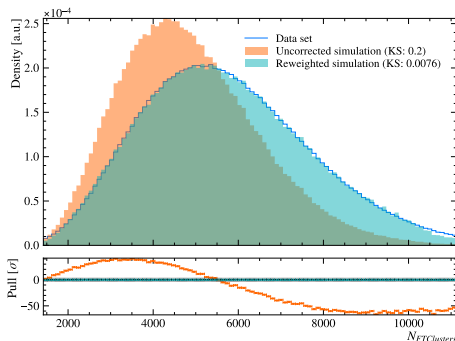
2026 Block 4 shows some discrepancies in Velo efficiencies

Multiplicity reweighting



- By default, reweight $nLongTracks$ and $nPVs$
- Need to change from unbinned to binned fit in TrackCalib2 to get correct uncertainties
- Reweighting code in TrackCalib2 by Maurice, but with XGBoost instead of GBReweigher

Alternative multiplicity reweighting



- Reweight `nFTClusters` as an alternative
- Assign systematic due to discrepancies in multiplicity between data and MC
- Weights obtained from simple histogram

Next steps:

- ① Produce correction tables for 2026 with expected MC?
- ② New iteration of 2024 tracking efficiencies with:
 - $\mu^\pm$, μ^+ , μ^- corrections
 - New 2D binning scheme
 - Full set of systematics
- ③ Look into block 1b MC when MC production is finished